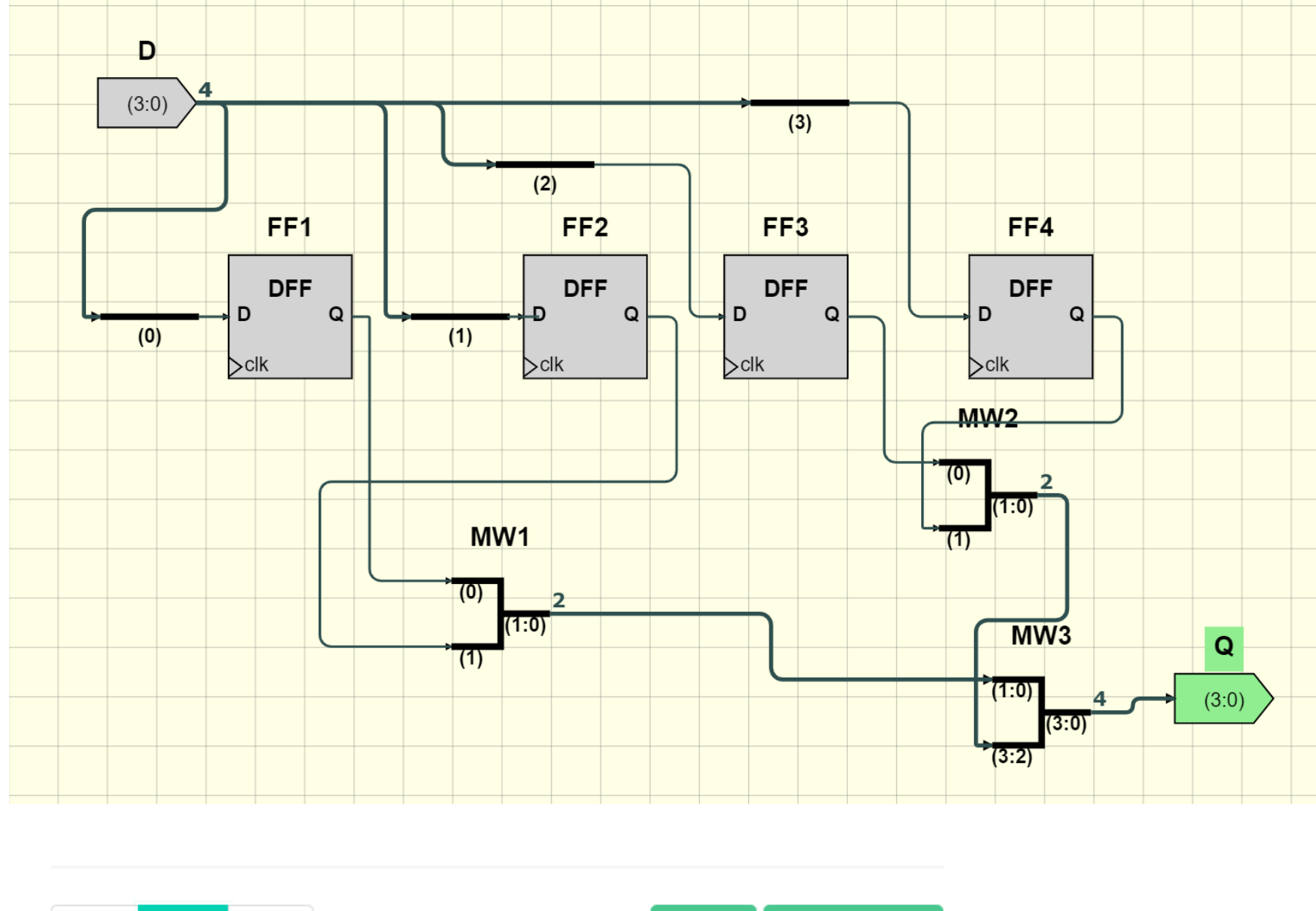


DECA 1.3 - Sequential circuits

Sunday, December 11, 2022

4:02 PM

Part 1 - Saving a register with D-type flip flops



0

1

0

1

Inputs

D (4 bits)

0

Outputs & Viewers

Q (4 bits)

0

Stateful components

DFF: FF4

0

DFF: FF3

0

DFF: FF2

0

DFF: FF1

0

Q is 0 as D has not been set
 If we go to a different clock tick, it stays at 0:

hex

dec

bin

Goto

Clock Tick 4

Inputs

D (4 bits)

0

Outputs & Viewers

Q (4 bits)

0

Stateful components

DFF: FF4

0

DFF: FF3

0

DFF: FF2

0

DFF: FF1

0

Now if I set D to a certain value, Q will remain 0:

hex	dec	bin
00	0	00000000
01	1	00000001
02	2	00000010
03	3	00000011
04	4	00000100
05	5	00000101
06	6	00000110
07	7	00000111
08	8	00001000
09	9	00001001
0A	10	00001010
0B	11	00001011
0C	12	00001100
0D	13	00001101
0E	14	00001110
0F	15	00001111
10	16	00010000
11	17	00010001
12	18	00010010
13	19	00010011
14	20	00010100
15	21	00010101
16	22	00010110
17	23	00010111
18	24	00011000
19	25	00011001
1A	26	00011010
1B	27	00011011
1C	28	00011100
1D	29	00011101
1E	30	00011110
1F	31	00011111
20	32	00100000
21	33	00100001
22	34	00100010
23	35	00100011
24	36	00100100
25	37	00100101
26	38	00100110
27	39	00100111
28	40	00101000
29	41	00101001
2A	42	00101010
2B	43	00101011
2C	44	00101100
2D	45	00101101
2E	46	00101110
2F	47	00101111
30	48	00110000
31	49	00110001
32	50	00110010
33	51	00110011
34	52	00110100
35	53	00110101
36	54	00110110
37	55	00110111
38	56	00111000
39	57	00111001
3A	58	00111010
3B	59	00111011
3C	60	00111100
3D	61	00111101
3E	62	00111110
3F	63	00111111
40	64	01000000
41	65	01000001
42	66	01000010
43	67	01000011
44	68	01000100
45	69	01000101
46	70	01000110
47	71	01000111
48	72	01001000
49	73	01001001
4A	74	01001010
4B	75	01001011
4C	76	01001100
4D	77	01001101
4E	78	01001110
4F	79	01001111
50	80	01010000
51	81	01010001
52	82	01010010
53	83	01010011
54	84	01010100
55	85	01010101
56	86	01010110
57	87	01010111
58	88	01011000
59	89	01011001
5A	90	01011010
5B	91	01011011
5C	92	01011100
5D	93	01011101
5E	94	01011110
5F	95	01011111
60	96	01100000
61	97	01100001
62	98	01100010
63	99	01100011
64	100	01100100
65	101	01100101
66	102	01100110
67	103	01100111
68	104	01101000
69	105	01101001
6A	106	01101010
6B	107	01101011
6C	108	01101100
6D	109	

Inputs

D (4 bits)

Outputs & Viewers

Q (4 bits)

Stateful components

DFF: FF4

DFF: FF3

DFF: FF2

DFF: FF1

...until the next clock tick:

100%

D (4 bits)

7

Outputs & Viewers

Q (4 bits)

7

Stateful components

DFF: FF4

0

DFF: FF3

1

DFF: FF2

1

DFF: FF1

1

And it will hold this value until I change D again:

hexdecbin

Goto

Clock Tick 9

D (4 bits)

Q (4 bits)

7

Stateful components

DFF: FF4

0

DFF: FF3

1

DFF: FF2

1

DFF: FF1

1

hexdecbin

Goto

Clock Tick 11

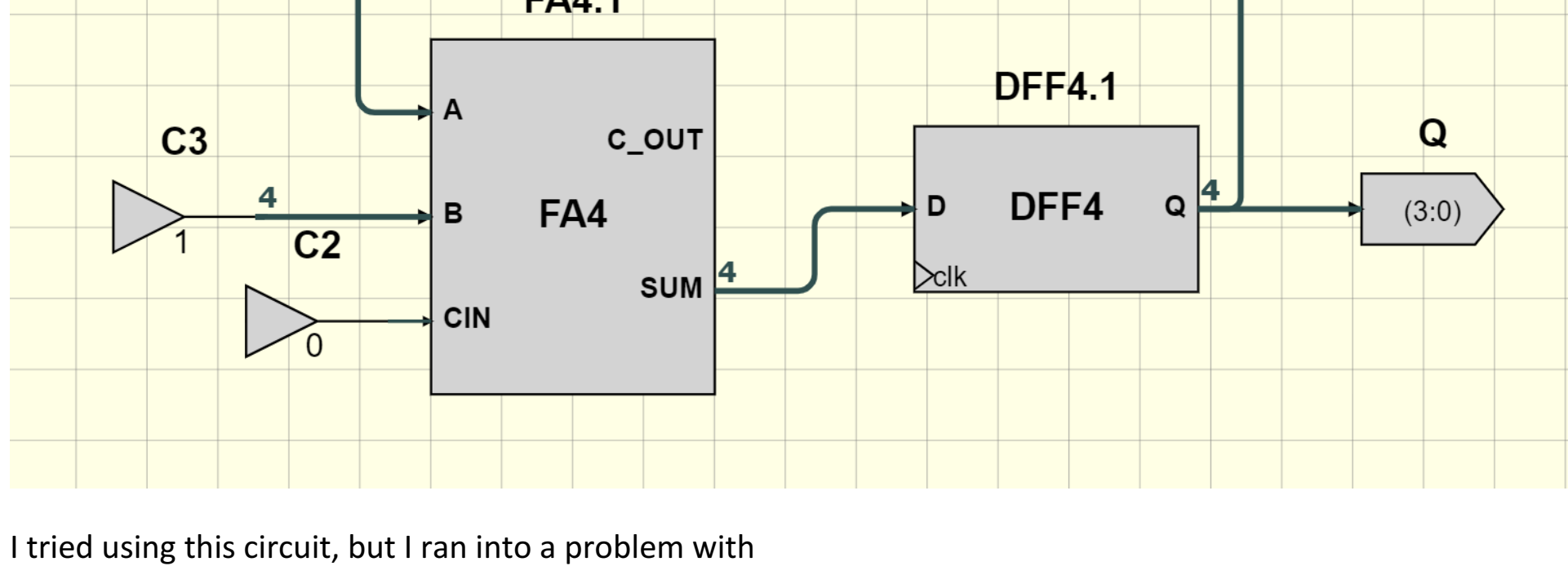
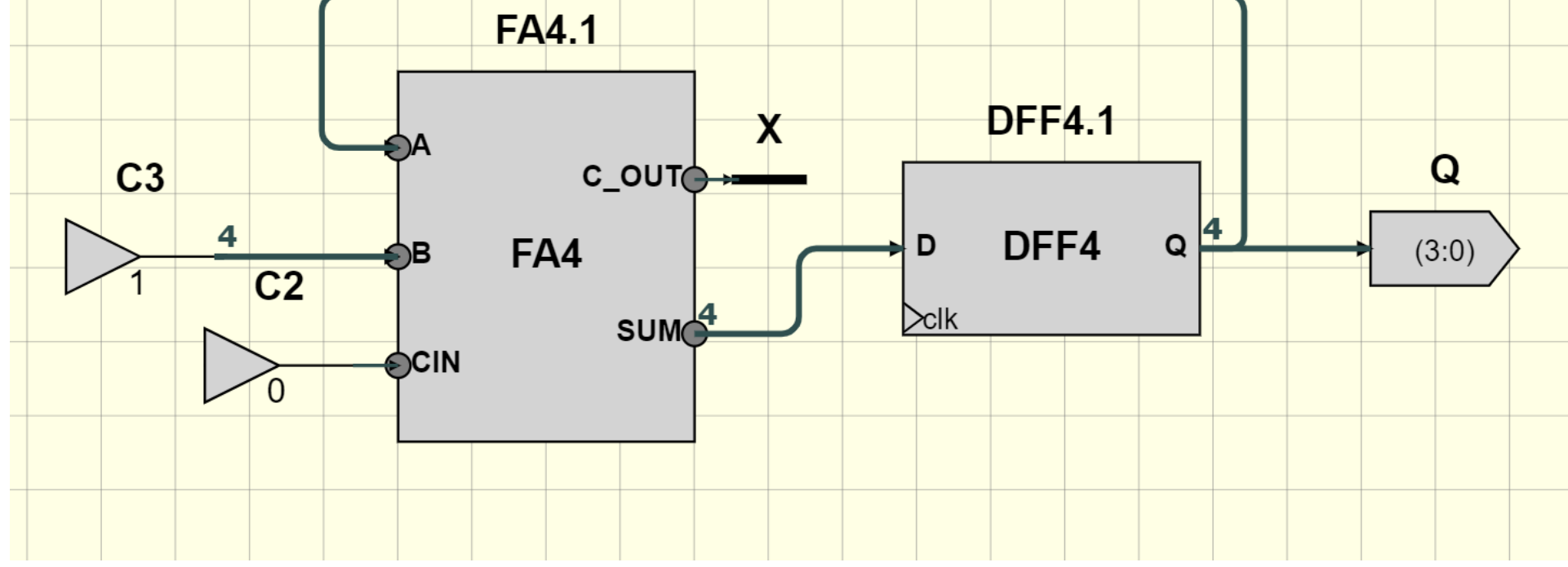
Inputs

D (4 bits)

2

Outputs & Viewers

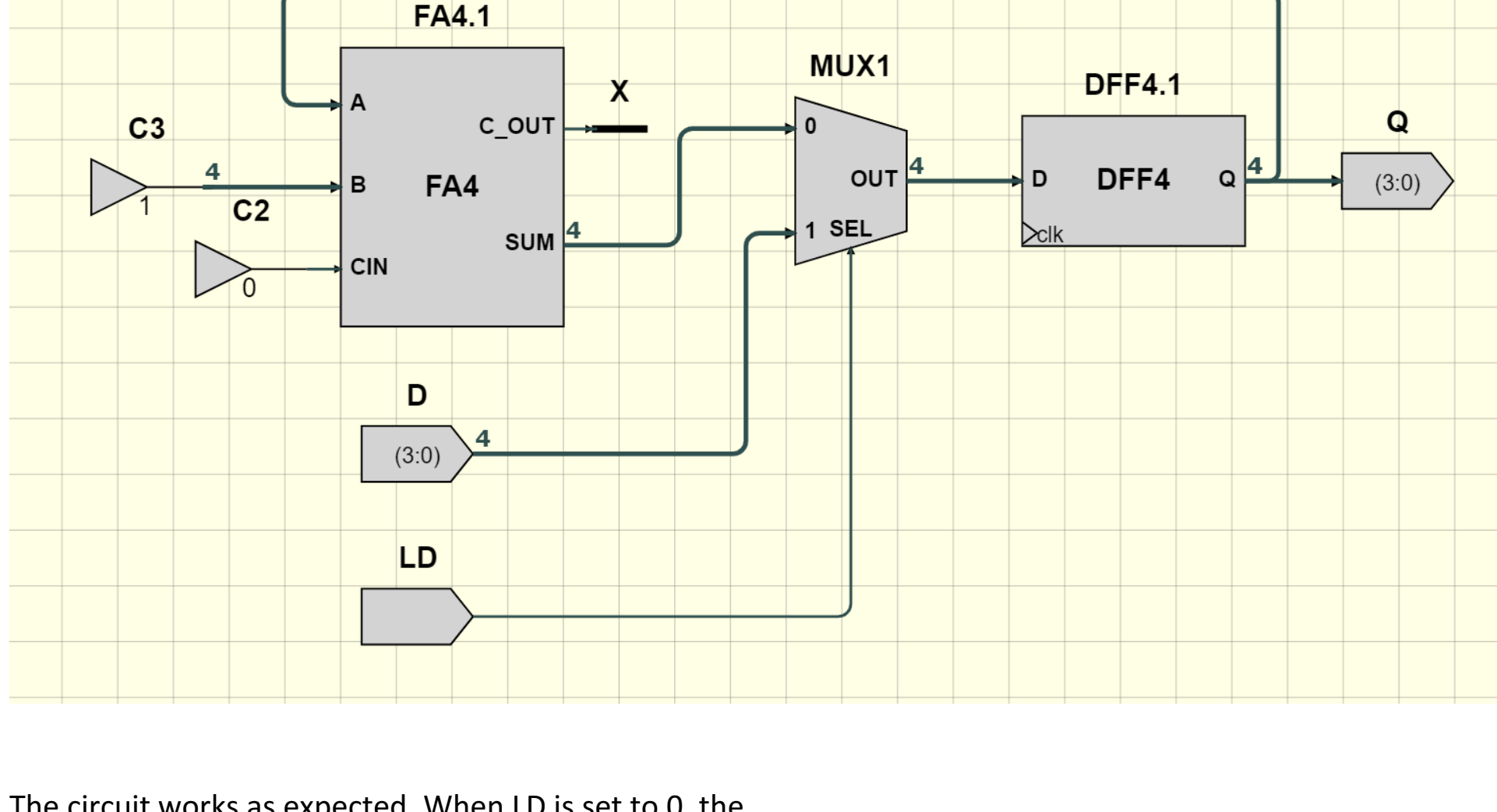
The diagram shows a 4-bit counter with four stateful components (DFFs) labeled DFF: FF4, DFF: FF3, DFF: FF2, and DFF: FF1. Each component has a data input and a data output. The outputs are connected in a chain: DFF: FF4 output to DFF: FF3 input, DFF: FF3 output to DFF: FF2 input, and DFF: FF2 output to DFF: FF1 input. The output of DFF: FF1 is labeled 'Count'. The current values of the outputs are shown in boxes to the right of each component: DFF: FF4 is 0, DFF: FF3 is 0, DFF: FF2 is 1 (highlighted in green), and DFF: FF1 is 0.

[illegible]

To fix this I connected it to a wire label named X for "don't care".

up from 0 to 15, then resetting back to 0.

Part 3: Counter with synchronous load

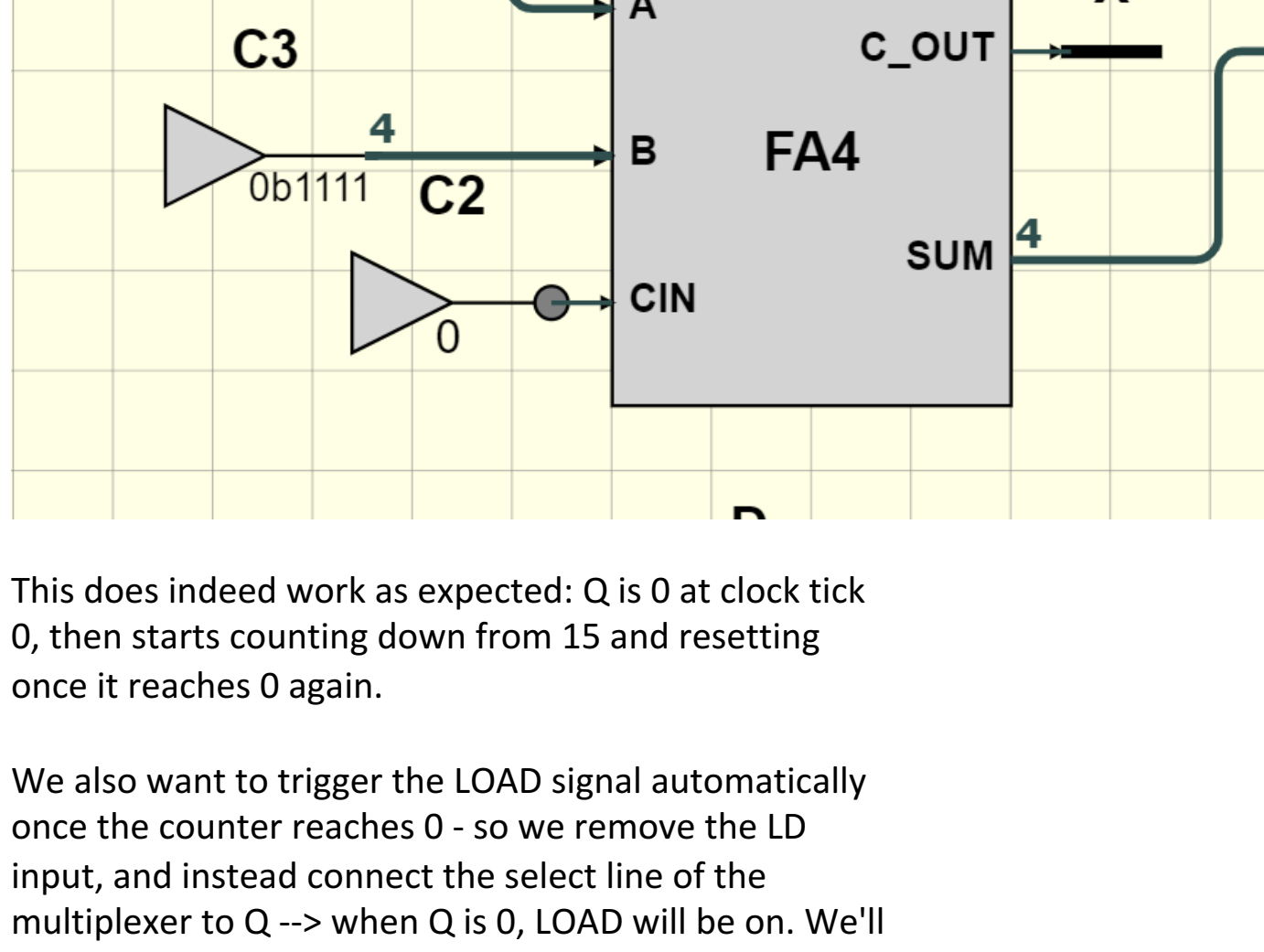
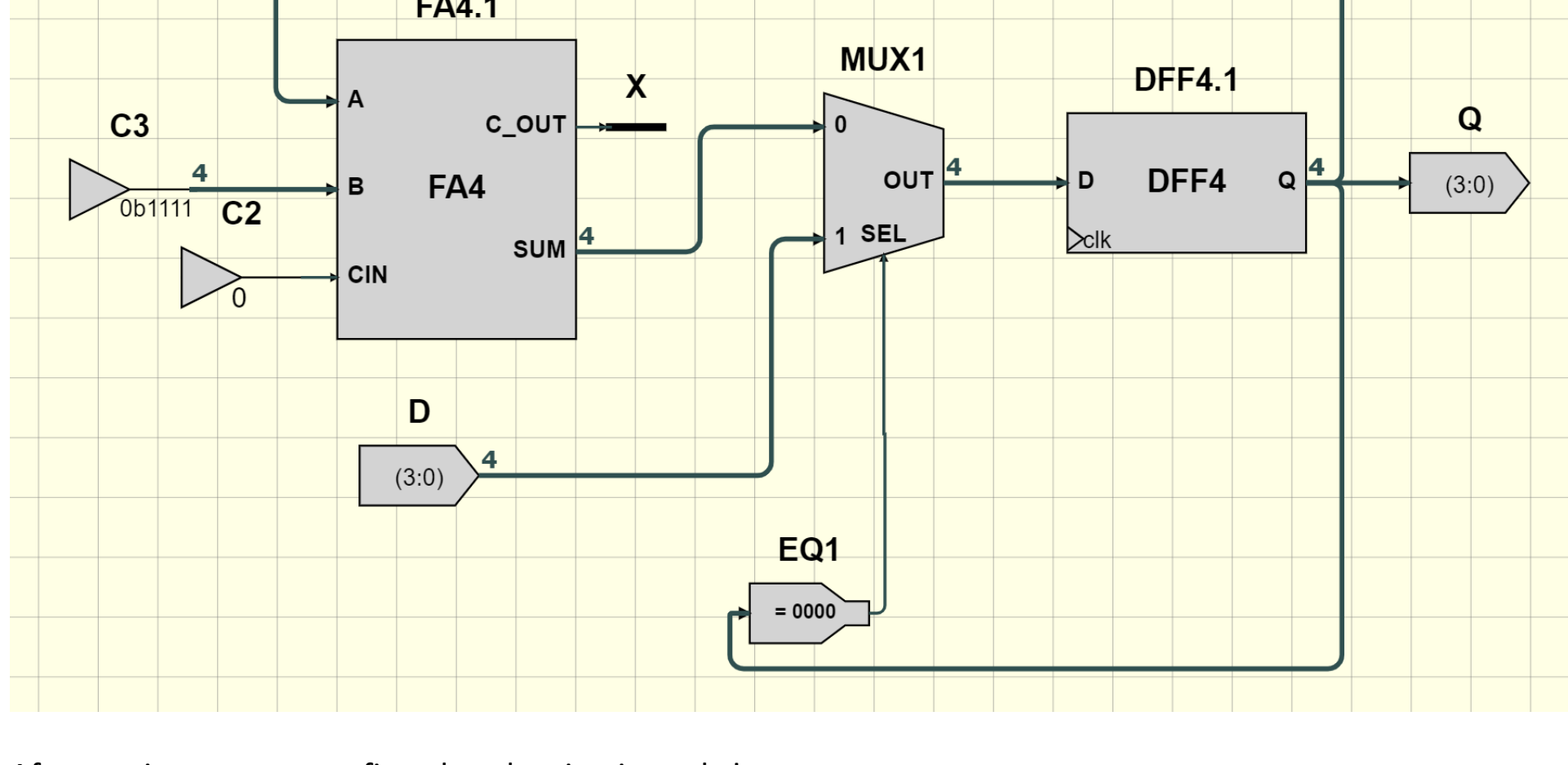


Part 4: Challenge - Countdown timer with auto-repeat

First we change the adder to a subtractor by setting it to 1111.

Let's first test if this works as expected before we

FA4.1

[illegible]

The counter counts down, and every time it reaches

it resets to the value of D.